

# Property Features or Host Effort?

## Inferring Superhost Status in the Melbourne Short-Term Rental Market

Group W06G02 · COMP20008 Elements of Data Processing, Assignment 2

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### 1. Introduction

*Research question.* Airbnb awards a ‘superhost’ status to hosts who meet a set of criteria. The criteria are at least 10 completed reservations (or 3 reservations totalling at least 100 nights), an overall rating of 4.8 or higher, a 90% response rate to new messages within 24 hours, and a cancellation rate below 1%, subject to specified exceptions (Airbnb, n.d.). Each criterion is either excluded from the analysis or unobservable in the data: the reservation and rating criteria correspond to the review-count columns and the review\_scores\_rating column with its sub-scores, all of which are excluded; the response-rate columns exist but are entirely empty, so response rate cannot be observed; and the dataset contains no column for cancellations.

The status signals exceptional service to guests and gives the listing more visibility, so hosts have a clear incentive to earn and keep it. We use the Inside Airbnb Melbourne dataset (Inside Airbnb, 2026), containing 25,728 listings described by 90 columns; the days\_since\_last\_review reference date is 2026-06-28, matching the latest last\_review in the data.

### 2. Methodology

#### 2.1. Data Preprocessing

We proposed and considered six candidate preprocessing steps (Table 1) for the Melbourne Airbnb dataset, aligned with the research question.

Table 1: Candidate preprocessing steps with dataset properties that supported the selection condition, with the corresponding outcome.

| # | Candidate step                                                                        | Properties                                                                                                       | Decision |
|---|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------|
| 1 | Deriving days_since_last_review from the raw last_review field                        | 4,477 rows (17.4%) have no review date at all, and their superhost rate is 6.59%                                 | Selected |
| 2 | Counting amenity_count by parsing the structured amenities field                      | Superhosts average 44.08 amenities, non-superhosts 31.03                                                         | Selected |
| 3 | Imputing missing price values via median imputation plus a missing indicator variable | 6,553 rows (25.47%) have no price                                                                                | Selected |
| 4 | Imputing missing values in the bedrooms and beds fields                               | Missing 18.19% for bedrooms and 27.44% for beds, and the missingness is conditional on room type                 | Rejected |
| 5 | Merging rare categories for the property_type variable                                | Of the 82 categories, 41 contain fewer than 10 listings, and these 41 together account for only 111 rows (0.43%) | Rejected |
| 6 | Splitting minimum_nights into short/long groups at a 30-night threshold               | Only 426 rows (1.66%) require 30 nights or more                                                                  | Rejected |

*Justifications for selections:* Deriving days\_since\_last\_review from the raw last\_review field was selected as dropping the 4,477 listings with no review date (17.4% of all rows) would raise the superhost rate from 30.65% to 35.72%. Furthermore, 295 superhost listings are still in the no review date group, suggesting the rows are worth keeping.

Amenities are also among the few features a host can change at any time, which places this column on the operational side of the research question.

Handling missing price values via imputation was selected as price missingness depends on whether the listing is bookable at all and is missing at random (MAR, not missing completely at random). 99.85% of unavailable listings (availability\_365 = 0) have no price, against 2.22% of available ones. As unavailable listings have an 8.13% superhost rate against 37.68% for available listings, deletion would inflate the superhost rate from 30.65% to 38.06%.

*Justifications for rejections:* Imputing the bedrooms and beds fields was rejected because missingness is structural by room type. Bedrooms are missing for 18.19% (beds 27.44%), but the data is split by room type, with entire homes missing 2.47%, private rooms 61.28%. Neither field enters the models in any case, since listing capacity is already represented by accommodates.

Merging property\_types was rejected because although there are 41 property types having fewer than 10 listings (out of 82 total property types), these 41 account for only 111 rows (0.43%), which is a trivial change to the data. The property\_type field is also excluded from the models, which use room\_type\_3 with three categories.

Splitting minimum\_nights into short/long stay bins was rejected as only 426 listings (1.66%) require a minimum stay of 30 nights or more. The long-stay group would be too small for the model to learn from.

## 2.2. Correlation Analysis

*Variable set:* The research question asks whether superhost status is more closely associated with listing fundamentals or host operational strategy, so the exploratory variable set must include representatives on both sides.

The five variables selected are host\_listings\_count, days\_since\_last\_review, availability\_365, dist\_cbd, and the binary target y ( $y = 1$  when host\_is\_superhost = 't'). Five variables give 10 unique pairs and 40 values.

*Rationale behind variable set:* On the listing fundamentals side, dist\_cbd measures the distance from Melbourne CBD (Flinders Street Station) and is appropriate as a fixed physical attribute that no host can change.

Price is excluded because 25.47% of prices are missing and 93.35% of those missing rows are unavailable listings, so keeping only priced rows would give a non-random sample. On the fundamentals side, accommodates has the lowest mutual information among the 11 features (0.0065), so dist\_cbd is used as the single fundamentals representative.

On the operational strategy side, the three variables selected characterise hosts based on three dimensions. host\_listings\_count captures scale (median 2, maximum 667 platform-wide; the largest host in this dataset has 319 listings), availability\_365 reflects calendar positioning (21.8% of listings at 0 days, 1.56% at 365) and days\_since\_last\_review measures recent activity (median 84 days, maximum 4,593). The variables are tilted 3:1 towards operational features: only dist\_cbd represents fundamentals, so the fair comparison between the two families relies on the feature-subset (A/B) experiments rather than this variable set.

*Computation:* The four exploratory variables are continuous and the target is binary, but no single coefficient is equally suitable for every pair. Applied to a 0/1 target, Pearson is a point-biserial correlation and reflects only the difference in means between the two groups, rather than the shape of the relationship. Spearman leaves the target with just two ranks and many ties, while MI and NMI require prior discretisation and therefore depend on how the bins are chosen. For MI and NMI, the continuous variables are binned using equal-frequency bins. We found  $n=10$  bins to be a good parameter: each bin holds roughly 2,125 rows, and the expected count in each cell of a  $10 \times 10$  contingency table is around 212, large enough to be reliable. Equal-width binning was rejected as 10 equal-width bins would concentrate 91.4% of host\_listings\_count rows in one bin (84.6% at 20 bins), which lowers MI. Correlations were only computed on 21,251 listings with at least one review, because 4,477 zero-review listings carry the imputation fill value 4,594 for days\_since\_last\_review, which would distort the coefficients because they all share the same artificial maximum value.

## 2.3. Supervised Learning Models and Evaluation

*Data:* All 25,728 listings are used, with zero-review listings marked by has\_review. The label is a host-level attribute, so every listing held by the same host carries the same label. Of 14,113 hosts, 3,770 are superhosts (26.71%), whereas superhost listings make up 30.65% of all listings. Because the label belongs to the host, the data must be split by host groups.

*Target and features:* The 11 features comprise three variables related to fundamentals (room\_type\_3, dist\_cbd, accommodates) and eight related to operational features (availability\_365, amenity\_count,

days\_since\_last\_review, has\_review, price\_num, price\_missing, minimum\_nights, host\_listings\_count). room\_type\_3 merges the 231 shared-room and 52 hotel-room rows (283 in total, 1.1%) into Other, leaving three categories that are then one-hot encoded. price\_num and price\_missing are treated as operational because the host sets the price, although price\_num and accommodates have a Spearman correlation of 0.682. This classification does not drive the result: moving price into the fundamentals group leaves its CV F1 at 0.1844 for the decision tree and 0.2647 for k-NN, against 0.5577 and 0.5442 for the operational group, so the conclusion is unchanged.

*Test/train split:* We use StratifiedGroupKFold, grouping by host\_id and stratifying by y, and take one fold as the test set, giving an 80/20 split. The 20,582 training rows contain 6,308 superhost listings and 14,274 non-superhost listings from 11,292 hosts, while the 5,146 test rows contain 1,577 and 3,569 listings from 2,821 hosts, with no host on both sides. The superhost share is held at 30.65% on each side. Grouping is necessary because the label belongs to the host and 55.8% of listings share a host with at least one other listing; under a random split, a model could score well simply by recognising a host it had already seen.

*Validation strategy:* Hyperparameters are tuned on the 20,582 training rows using 5-fold, host-grouped and stratified cross-validation, and each fold is scored by F1 for the superhost class. Each validation fold holds about 4,116–4,117 rows, including 1,261–1,262 superhost listings from 2,258–2,259 hosts. Ten folds were not used because the largest host holds 319 listings; that host would occupy 7.75% of a validation fold under five folds and 15.5% under ten, so a single host could dominate an entire fold.

*Hyperparameter grids:* k-NN is trained with  $k \in \{1, 5, 11, 21, 51, 101, 151, 201, 301\}$  and the decision tree with  $\text{max\_depth} \in \{2, 3, 4, 5, 6, 8, 10, 15, \text{None}\}$ , where None means unlimited depth and shows the cost of memorising the training set (0.5018). Only odd values of k are tested so that a binary vote cannot end in a tie, the split criterion stays at its default of gini, and for each model the value with the best mean cross-validated F1 is selected.

*Baseline and metrics:* The primary metric is the F1 score of the superhost class, and per-class precision and recall and overall accuracy are reported alongside. The baseline is a majority-class classifier, which predicts “not a superhost” for every listing; it reaches 0.694 accuracy and an F1 score of exactly 0. A percentile bootstrap with 1,000 resamples drawn with replacement gives a 95% confidence interval for test F1. The test set contains 2,821 hosts and 5,146 rows, and listings of the same host share a label, so resampling rows would overstate the number of independent units and hosts are resampled instead. The two models are scored together on every resample, so the procedure directly gives an interval for their difference. Repeated cross-validation is not used because the target is the uncertainty in the single reported test score, not the variation across different training sets.

*Feature-set experiments:* To answer the RQ, both models are retrained on six labelled feature sets. A uses the three fundamentals features, B the eight operational features, C all eleven, D removes the review pair from C, E removes availability\_365 from C, and F removes the review pair from B; two further checks, G (C without amenity\_count) and H (C without the price pair), support the preprocessing discussion. Each set uses the hyperparameters selected in Table 5 and is scored by cross-validation on the training rows, with the test score used only for confirmation.

#### 2.4. Feature Selection

*Filter method:* Mutual information with the target is estimated independently for each feature on the training set using scikit-learn’s mutual\_info\_classif function. Room type is integer encoded, the discrete features (room\_type\_3, has\_review, price\_missing) are declared, and remaining missing values are filled with the training median. These continuous-value MIs are not comparable to the binned MI in Table 4.

*Embedded method:* Impurity-based importance is taken from the tuned depth-10 decision tree, measured by Gini impurity, and the tree is trained on the training set only.

#### 2.5. Dimensionality Reduction & Clustering

*Candidate feature sets.* Clustering runs on the 21,251 reviewed listings, since the 4,477 zero-review listings share the imputed value 4,594, producing a separate cluster. Three candidate feature sets are compared quantitatively in Table 2: (S1) operational only, (S2) fundamentals only, and (S3) all six features.

Three of the six features are log-transformed via  $\log_{1p}$ , and all six were standardised via StandardScaler. Without log transform a single host with 667 listings would dominate a component on its own.

Table 2: Candidate feature sets for clustering, built from the two families in the research question. Every candidate set follows the same pipeline of  $\log_{1p}$  transformation, standardisation and K-Means with  $k = 5$ : the superhost-rate range is the difference between the highest and lowest cluster superhost rates the silhouette score is computed on a 5,000-row sample, and the smallest cluster is the number of listings in the least populated cluster.

| Set | Name                  | Features                                                                                             | Superhost-rate range | Silhouette | Smallest cluster |
|-----|-----------------------|------------------------------------------------------------------------------------------------------|----------------------|------------|------------------|
| S1  | Operational only      | availability_365, amenity_count, days_since_last_review, host_listings_count                         | 0.5331               | 0.2942     | 3,243            |
| S2  | Fundamentals only     | dist_cbd, accommodates                                                                               | 0.0754               | 0.3997     | 780              |
| S3  | Mixed (both families) | availability_365, amenity_count, days_since_last_review, host_listings_count, dist_cbd, accommodates | 0.4719               | 0.2232     | 2,365            |

*Selected feature set.* Feature set S3 is used because the research question asks whether superhost status is closer to listing fundamentals or to operational strategy, and a set holding only one family cannot answer that. S1 separates slightly better but contains operational features alone, so it cannot test which family matters, whereas S3 lets both families compete while retaining most of S1’s separation (a superhost-rate range of 0.4719 versus 0.5331). The very small range for S2 (0.0754) is itself informative, because clustering on fundamentals alone produces groups that barely differ in superhost status.

*Clustering and PCA.*  $k=5$  is read from the bend in the SSE curve: the SSE reduction when adding each cluster is 16.0%, 12.8%, 12.0% and 6.8% for  $k=2 \rightarrow 3$ ,  $3 \rightarrow 4$ ,  $4 \rightarrow 5$  and  $5 \rightarrow 6$ . The silhouette score is also highest at  $k=5$  (0.2232, versus 0.2227 at  $k=4$  and 0.2177 at  $k=6$ ). For hierarchical clustering, Ward linkage with Euclidean distance is used at the same  $k=5$ . The three linkages presented in the course all collapse, on these features at  $k=5$ , into one dominant cluster: single linkage gives sizes [21,247, 1, 1, 1, 1], average linkage [19,482, 1,401, 341, 26, 1], and complete linkage a largest cluster of 12,473. Ward instead merges the two clusters that add the least to the within-cluster SSE, which is the same objective that K-Means minimises, so the methods can be compared directly. PCA is applied to feature set S3, retaining all six components so that the explained variance of the first three can be reported.

### 3. Results Exploration and Analysis

#### 3.1. Data Preprocessing

Table 3: Measurable before/after impact of the three selected preprocessing steps.

| Candidate step                                                   | Before                                                                                | After                                                                                                                            |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Deriving days_since_last_review from last_review                 | last_review missing for 4,477 rows, could not derive missing values                   | days_since_last_review filled with 4,594 days for missing last_review rows. 0 missing rows with median 84 days.                  |
| Counting amenity_count by parsing the structured amenities field | No field exists describing the amenity count of a particular listing.                 | A naive comma count would be wrong on 3,096 rows (12.03%), over-counting by up to 21 amenities for a single listing.             |
| Imputing missing price values via median                         | 25.47% of prices missing; the missingness depends on availability_365 (MAR, not MCAR) | 0% prices missing, imputed within the pipeline using the training-set median of \$245.00. price_missing marks every imputed row. |

#### 3.2. Correlation Analysis

Pearson and Spearman disagree on host\_listings\_count with the target, at  $-0.069$  and  $+0.070$ , because the superhost rate does not move in one direction. This pattern is shown in Figure 1.

#### 3.3. Supervised Learning Models and Evaluation

#### 3.4. Feature Selection

*Top features from each method.* Table 9 reports the full feature ranking under mutual information and under tree impurity, with the corresponding values.

Table 4: Pearson, Spearman, mutual information (bits) and normalised mutual information for each pair of the five variables (n=21,251 listings with at least one review). Rows are ordered by NMI, highest first.

| Pair                                         | Pearson | Spearman | MI (bits) | NMI    | Bins  |
|----------------------------------------------|---------|----------|-----------|--------|-------|
| days_since_last_review ~ availability_365    | -0.438  | -0.395   | 0.2639    | 0.0856 | 10×8  |
| days_since_last_review ~ y                   | -0.337  | -0.443   | 0.1672    | 0.0785 | 10×2  |
| availability_365 ~ y                         | 0.146   | 0.155    | 0.0613    | 0.0324 | 8×2   |
| host_listings_count ~ dist_cbd               | -0.131  | -0.225   | 0.0837    | 0.0285 | 7×10  |
| host_listings_count ~ days_since_last_review | -0.113  | -0.194   | 0.0806    | 0.0274 | 7×10  |
| host_listings_count ~ availability_365       | 0.059   | 0.225    | 0.0621    | 0.0230 | 7×8   |
| availability_365 ~ dist_cbd                  | 0.170   | 0.124    | 0.0455    | 0.0147 | 8×10  |
| days_since_last_review ~ dist_cbd            | -0.081  | 0.061    | 0.0458    | 0.0138 | 10×10 |
| host_listings_count ~ y                      | -0.069  | 0.070    | 0.0169    | 0.0097 | 7×2   |
| dist_cbd ~ y                                 | 0.076   | 0.009    | 0.0101    | 0.0047 | 10×2  |

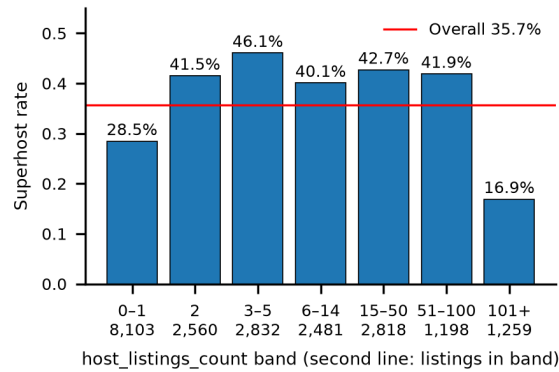


Figure 1: Superhost rate by host\_listings\_count band (n = 21,251 reviewed listings). The red line marks the overall superhost rate of 35.7%.

Table 5: Cross-validated F1 score at every hyperparameter value tried (five grouped folds, 20,582 training listings). Selected configurations shaded blue. Values are mean  $\pm$  standard deviation of F1 across the five folds.

|                        |        |        |        |        |        |               |               |        |        |
|------------------------|--------|--------|--------|--------|--------|---------------|---------------|--------|--------|
| <b>k-NN: k</b>         | 1      | 5      | 11     | 21     | 51     | <b>101</b>    | 151           | 201    | 301    |
| CV F1 (mean)           | 0.4802 | 0.5020 | 0.5071 | 0.5190 | 0.5299 | <b>0.5370</b> | 0.5312        | 0.5309 | 0.5249 |
| $\pm$ SD               | 0.0175 | 0.0159 | 0.0176 | 0.0112 | 0.0121 | 0.0133        | 0.0147        | 0.0175 | 0.0226 |
| <b>Tree: max_depth</b> | 2      | 3      | 4      | 5      | 6      | 8             | <b>10</b>     | 15     | None   |
| CV F1 (mean)           | 0.5158 | 0.5611 | 0.5712 | 0.5384 | 0.5512 | 0.5622        | <b>0.5745</b> | 0.5363 | 0.5018 |
| $\pm$ SD               | 0.0503 | 0.0431 | 0.0177 | 0.0424 | 0.0240 | 0.0123        | 0.0130        | 0.0171 | 0.0072 |

Table 6: Change in mean 5-fold CV F1 when each tuned hyperparameter is moved from its default to the selected value.

| Model         | Hyperparameter                    | Default          | Chosen         | Default CV F1 | Chosen CV F1 | Effect  |
|---------------|-----------------------------------|------------------|----------------|---------------|--------------|---------|
| k-NN          | Number of neighbours (k)          | k = 5            | k = 101        | 0.5020        | 0.5370       | +0.0350 |
| Decision Tree | Maximum depth of tree (max_depth) | max_depth = None | max_depth = 10 | 0.5018        | 0.5745       | +0.0727 |

Table 7: Test-set performance (n=5,146), with precision, recall, accuracy and F1 scores.

| Model                    | Class         | Precision | Recall | Accuracy | F1 Score |
|--------------------------|---------------|-----------|--------|----------|----------|
| Baseline                 | Non-superhost | 0.6935    | 1.0000 | 0.6935   | 0.8190   |
|                          | Superhost     | 0.0000    | 0.0000 |          | 0.0000   |
| k-NN (k=101)             | Non-superhost | 0.8106    | 0.8322 | 0.7487   | 0.8212   |
|                          | Superhost     | 0.5958    | 0.5599 |          | 0.5773   |
| Decision Tree (depth=10) | Non-superhost | 0.8445    | 0.8067 | 0.7629   | 0.8252   |
|                          | Superhost     | 0.6028    | 0.6639 |          | 0.6319   |

Table 8: 95% percentile-bootstrap intervals (1,000 paired host resamples) for test-set superhost-class F1, for the gap between models and for feature-subset comparisons. A, B and F are decision-tree results.

| Model/Pair                      | 95% CI           | Point estimate |
|---------------------------------|------------------|----------------|
| k-NN                            | [0.5423, 0.6155] | 0.5773         |
| Decision Tree                   | [0.5940, 0.6757] | 0.6319         |
| DT – k-NN                       | [0.0130, 0.0986] | 0.0546         |
| A (fundamentals)                | [0.1535, 0.2384] | 0.1955         |
| B (operational)                 | [0.5981, 0.6844] | 0.6389         |
| B – A                           | [0.3813, 0.5046] | 0.4434         |
| F (operational, no review pair) | [0.4923, 0.6313] | 0.5662         |
| F – A                           | [0.2819, 0.4536] | 0.3707         |

*Hard case.* The tree’s most confident miss is listing id 6812677, belonging to host 35667203, who has only this one listing; it is a superhost listing that the tree classified as a non-superhost with a probability of 0.000. It is one of 530 false negatives.

Table 9: Full ranking of the 11 features by mutual information and decision-tree impurity, with permutation importance for both models.

| Feature                | MI     | MI rank | Impurity | Impurity rank | DT permutation | k-NN permutation |
|------------------------|--------|---------|----------|---------------|----------------|------------------|
| days_since_last_review | 0.1273 | 1       | 0.5005   | 1             | 0.2269         | 0.0747           |
| host_listings_count    | 0.1087 | 2       | 0.2698   | 2             | 0.0195         | 0.0037           |
| availability_365       | 0.0759 | 3       | 0.0356   | 4             | 0.0131         | 0.0169           |
| amenity_count          | 0.0749 | 4       | 0.1023   | 3             | 0.0810         | 0.1433           |
| price_num              | 0.0657 | 5       | 0.0277   | 6             | 0.0109         | 0.0039           |
| price_missing          | 0.0451 | 6       | 0.0024   | 9             | 0.0219         | 0.0621           |
| has_review             | 0.0344 | 7       | 0.0000   | 11            | 0.0000         | 0.0073           |
| dist_cbd               | 0.0340 | 8       | 0.0278   | 5             | 0.0054         | 0.0324           |
| minimum_nights         | 0.0097 | 9       | 0.0157   | 8             | 0.0014         | 0.0006           |
| room_type_3            | 0.0089 | 10      | 0.0001   | 10            | 0.0000         | 0.0168           |
| accommodates           | 0.0065 | 11      | 0.0181   | 7             | 0.0062         | -0.0064          |

### 3.5. Dimensionality Reduction & Clustering

*PCA.* The first three components carry 69.4% of the variance between them, and Table 10 reports the explained variance and the top loadings. Figure 2 plots the listings against the first two components.

Table 10: PCA on standardised set S3, containing explained variance ratio (EVR), cumulative variance and top 3 loadings.

| Component | EVR    | Cumulative | Rank 1                        | Rank 2                     | Rank 3                  |
|-----------|--------|------------|-------------------------------|----------------------------|-------------------------|
| PC1       | 0.3211 | 0.3211     | days_since_last_review +0.563 | amenity_count –0.504       | availability_365 –0.447 |
| PC2       | 0.2045 | 0.5255     | dist_cbd +0.786               | host_listings_count –0.556 | accommodates +0.208     |
| PC3       | 0.1681 | 0.6936     | accommodates +0.671           | availability_365 –0.591    | amenity_count +0.355    |

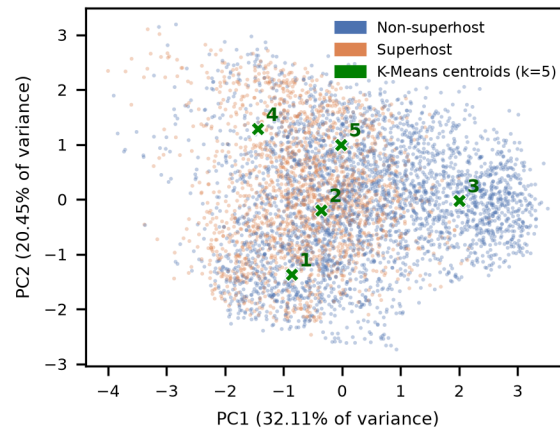


Figure 2: PCA scatter of the 21,251 reviewed listings on the first two components, coloured by superhost status, with K-Means centroids (k=5) marked. A random sample of 5,000 listings is shown. Centroids numbered 1–5 correspond to clusters K1–K5 in Table 11.

Table 11: K-Means (K1–K5) and Ward (W1–W5) cluster profiles (k=5) on the standardised six-feature set. Feature values are medians, in original units (km, guests, days, counts), not standardised values.

| #         | Listings | Share | Superhost<br>rate | dist_cbd | accom-<br>modates | availability<br>_365 | amenity<br>_count | days_since<br>_last<br>_review | host<br>_listings<br>_count |
|-----------|----------|-------|-------------------|----------|-------------------|----------------------|-------------------|--------------------------------|-----------------------------|
| <b>K1</b> | 4,541    | 21.4% | 35.3%             | 1.42     | 4                 | 188                  | 41                | 50                             | 50                          |
| <b>K2</b> | 5,340    | 25.1% | 53.9%             | 2.25     | 4                 | 157                  | 47                | 40                             | 2                           |
| <b>K3</b> | 4,689    | 22.1% | 6.8%              | 4.54     | 2                 | 0                    | 23                | 1,883                          | 1                           |
| <b>K4</b> | 2,365    | 11.1% | 43.7%             | 20.56    | 8                 | 218                  | 51                | 58                             | 3                           |
| <b>K5</b> | 4,316    | 20.3% | 40.7%             | 17.82    | 2                 | 310                  | 33                | 76                             | 2                           |
| <b>W1</b> | 8,256    | 38.9% | 49.9%             | 5.36     | 4                 | 174                  | 47                | 56                             | 1                           |
| <b>W2</b> | 3,829    | 18.0% | 30.1%             | 7.32     | 2                 | 267                  | 22                | 100                            | 3                           |
| <b>W3</b> | 4,716    | 22.2% | 37.5%             | 1.79     | 4                 | 175                  | 42                | 55                             | 45                          |
| <b>W4</b> | 3,249    | 15.3% | 2.9%              | 4.34     | 2                 | 0                    | 20                | 2,364                          | 1                           |
| <b>W5</b> | 1,201    | 5.7%  | 37.8%             | 17.55    | 10                | 221                  | 49                | 49                             | 12                          |

## 4. Discussion and Interpretation

### 4.1. Data Preprocessing

The three preprocessing steps have different effects on the downstream models, as shown by the feature-ablation experiments. Removing the review-related features (experiment D) moves the decision tree’s CV F1 from 0.5745 to 0.5262 (test F1 0.6319 to 0.5667) and k-NN from 0.5370 to 0.5032. Removing amenity\_count (experiment G) moves k-NN from 0.5370 to 0.4747 (test 0.5773 to 0.5196), while the decision tree is almost unchanged at 0.5747. Removing the price pair (experiment H) moves k-NN to 0.5220 and the decision tree slightly up to 0.5791. The decision tree therefore relies almost entirely on review recency, whereas k-NN also depends on amenity\_count and price, which matches the permutation-importance results in Table 9.

### 4.2. Correlation Analysis

*Agreement and divergence.* The two coefficients agree closely on the strongest pair, days\_since\_last\_review with availability\_365 (−0.438 vs −0.395), but they point in opposite directions on two pairs (Table 4).

The host\_listings\_count result can be explained from the ranks and the raw values. Spearman works with ranks: most listings fall in the 0–5 host-listings range (8,103, 2,560 and 2,832 listings across its three bands), where the superhost rate rises from 28.5% to 46.1%, so the rank correlation is positive at +0.070. Pearson uses the raw values: 1,259 listings belong to hosts with more than 100 listings (up to the platform-wide maximum of 667), and their superhost rate is only 16.9%, which pulls the coefficient to −0.069. The two coefficients therefore describe the same non-monotonic relationship but weight different parts of it; MI and NMI do not assume monotonicity, yet NMI for this pair is only 0.0097.

*The second sign reversal.* Between days\_since\_last\_review and dist\_cbd the same pattern appears (Pearson −0.081, Spearman +0.061). When distance is split into twenty equal-frequency bands, the median days\_since\_last\_review is 41–60 days at 0.6–1.4 km, rises to 111 days at 1.7 km, peaks at 122 days near 5.0 km, and then falls to 69 days at 34.6 km and 47 days at 50.6 km. Most listings lie between the centre and the middle suburbs, where recency rises with distance, giving the positive Spearman value, while a small number of distant listings with low recency pull the raw-scale Pearson value negative.

*Connection to the RQ.* Of the four pairs ranked by NMI, days\_since\_last\_review has the strongest association (NMI 0.0785), followed by availability\_365 (0.0324) and host\_listings\_count (0.0097). dist\_cbd comes last (0.0047), so the only listing fundamentals feature in the set has the weakest association with the target. It is important to note that these are associations only. Hosts with more than 100 listings may be running business or apartment models, so their low rate may reflect the business, rather than service quality. Not every guest also leaves a review, so days\_since\_last\_review may be inaccurate. A further confounder is that days\_since\_last\_review partly substitutes for an excluded criterion, since its Spearman correlation with number\_of\_reviews\_ltm is −0.8155. This motivates experiment F, which removes the review-related operational features; the test-set interval for F – A, [0.2819, 0.4536], remains clearly positive.

### 4.3. Supervised Learning Models and Evaluation

*Model performance.* With the official criteria excluded, the tree reaches a test F1 of 0.632 on the superhost class and k-NN 0.577, against a majority-class baseline of exactly 0. The baseline model that ignores every feature already reaches 0.694 accuracy, which is just the non-superhost share restated ( $1 - 0.3065 = 0.6935$ ), so accuracy cannot separate a useful model from the baseline. Relative to the baseline, F1 rises by 0.6319 for the tree and 0.5773 for k-NN, accuracy by 0.0694 (0.7629 versus 0.6935) and 0.0552, and macro-F1 by 0.3190 and 0.2898 (0.7285 and 0.6993 against 0.4095).

*Comparing the models.* The decision tree is the stronger of the two. Its paired bootstrap interval against k-NN is [0.0130, 0.0986], which does not include zero, so the difference is unlikely to be sampling noise. The tree labels more listings as superhosts, 1,737 in total against 1,482 for k-NN, which is a trade-off: of these, 1,047 and 883 are true positives, 690 and 599 are false positives.

*Feature influence.* The two models rely on different features. For the decision tree, permutation importance is highest for `days_since_last_review` (0.2269), followed by `amenity_count` (0.0810); for k-NN it is highest for `amenity_count` (0.1433), followed by `days_since_last_review` (0.0747) and `price_missing` (0.0621). `host_listings_count` shows a striking contrast, with impurity importance of 0.2698 (second highest) but permutation importance of only 0.0195 for the tree and 0.0037 for k-NN, meaning the tree often splits on it during training although it adds little to test predictions. The features that both models depend on most are operational, which is consistent with the RQ.

*Model behaviour and limits.* Two further points qualify the results. First, depth 10 gives a CV F1 of  $0.5745 \pm 0.0130$  and depth 4 gives  $0.5712 \pm 0.0177$ , a difference of only 0.0033 that is smaller than the standard deviation, so the two depths cannot be reliably separated. Depth 10 is retained because the selection rule, fixed before tuning began, was to take the configuration with the highest mean CV F1 as stated in Section 2.3; under a one-standard-error rule the simpler depth 4 would instead have been selected, and this is noted as a limitation in Section 5. Second, the decision tree shows mild overfitting, with training F1 of 0.7188 against test F1 of 0.6319 (a gap of 0.087), whereas k-NN has a smaller gap, 0.6023 against 0.5773, because  $k=101$  applies strong smoothing.

*Subset experiments and the RQ.* The subset experiments answer the research question most directly. On cross-validation, fundamentals reach 0.166 F1 and operational features 0.548; on the test set the corresponding values are 0.1955 and 0.6389, and the 95% interval for the test-set difference of 0.4434 is [0.3813, 0.5046], which does not include zero.

### 4.4. Feature Selection

*Agreements and divergence.* Both put `days_since_last_review` first and `host_listings_count` second, but they swap the next two features: `availability_365` ranks third by mutual information (0.0759) but fourth by impurity (0.0356), while `amenity_count` ranks fourth by mutual information (0.0749) but third by impurity (0.1023). Mutual information scores each feature against the target on its own, so the redundancy between `availability_365` and `days_since_last_review` is not penalised. Feature importance works differently: it credits a feature only for the tree impurity it removes once the earlier splits are already in place. `availability_365` carries much the same information as `days_since_last_review`, so once the tree has split on recency there is little left for it to explain, and it drops to 0.0356. `amenity_count` overlaps with recency far less, so it keeps 0.1023. Experiment E supports this reading: removing `availability_365` leaves CV F1 slightly higher (0.5846 vs 0.5745) and test F1 essentially unchanged (0.6303 vs 0.6319).

*The hard case.* Listing 6812677 is a superhost listing that the tree classed as a non-superhost with a predicted probability of 0.000. On the features the model can see it looks less active: 18 amenities against medians of 45 for superhosts and 32 for non-superhosts, `availability_365` of 116 which is below even the non-superhost median of 124, a host with a single listing, and 156 days since the last review against a superhost median of 37. The tree path explains the result: the root split is `days_since_last_review`  $\leq 147.5$ , and its value of 156 falls only 8.5 days onto the inactive side; it then meets `host_listings_count`  $\leq 1.5$  and `amenity_count`  $\leq 24.5$  (it has 18), ending in a leaf of 86 training listings that are all non-superhosts, hence the probability of 0.000. k-NN also classifies it incorrectly, with a probability of 0.267. On the two official criteria observable in the data, it meets both: 24 reviews in the last twelve months (at least 10 required) and a rating of 4.8 (at least 4.8 required), but these two features are excluded from the models.

Its `days_since_last_review` sits at the 54.6th percentile among all listings yet the 90.8th percentile among reviewed superhosts, so it is ordinary overall but extreme within its own class.

*Filter-high, embedded-low.* Two features show why a high filter ranking can coincide with a low embedded ranking. `has_review` has MI of 0.0344 (rank 7) but impurity of 0.0000 (rank 11), because `has_review = 0` is exactly equivalent to `days_since_last_review = 4,594`, and once the tree splits on recency it has captured all of that information. `price_missing` ranks sixth by MI but ninth by impurity, since 93.35% of listings with a missing price also have `availability_365 = 0`, so the two features largely overlap.

#### 4.5. Dimensionality Reduction & Clustering

*PCA.* PC1 carries 32.1% of the variance, and its three largest loadings are all operational: `days_since_last_review` (+0.563), `amenity_count` (−0.504) and `availability_365` (−0.447). PC2 carries 20.45% and compares `dist_cbd` (+0.786) against `host_listings_count` (−0.556), which separates distance from the CBD from the size of the host's portfolio. PC3 carries 16.8% and compares `accommodates` (+0.671) against `availability_365` (−0.591). The components also differ in how they relate to the target: PC1 has correlation −0.354 and NMI 0.057 with `y`, whereas PC2 has correlation +0.050 and NMI 0.0036. PC2 explains 20.45% of the variance along the geographic and scale axis but is almost unrelated to superhost status, so high variance does not imply importance for the RQ; the structure linked to superhost status lies almost entirely on PC1, the operational axis.

*K-Means.* K-Means separates two clusters at the extremes rather than one. K3 holds 22.1% of listings, with a median of 1,883 days since the last review, zero availability across the year, one listing, 23 amenities and a superhost rate of 6.8%; K2 sits at the opposite end, with 40 days since the last review, 47 amenities and a rate of 53.9%. K1 (35.3%) and K5 (40.7%) stay close to the 35.72% base rate and K4 (43.7%) sits slightly above it, and the three differ mainly in location and size: K1 is a group of large operators with a median of 50 listings 1.4 km from the CBD, K4 is larger homes at 20.6 km, and K5 is small listings at 17.8 km with 310 available days.

*Connection to the RQ.* PCA shows that the axis that explains the most is operational, which agrees with the supervised result that fundamentals alone reach 0.166 CV F1 against 0.548 for operational features.

*K-Means versus Ward.* Comparing K-Means with Ward shows the same structure with different boundaries. The adjusted Rand index is 0.377 and the majority overlap is 68.4%. The dormant cluster is stable under both methods: 3,224 of the 4,689 listings in K3 fall into W4, and 99.2% of W4's 3,249 listings come from K3. The large-operator cluster also corresponds closely, with 81.4% of K1 (3,696 listings) falling into W3, while 82.6% of K2 (4,410 listings) falls into W1. The Ward profiles complete the interpretation: W1 (8,256 listings, 49.9% superhosts, 47 amenities, 56 days since review), W2 (3,829, 30.1%, 22 amenities, 267 available days), W3 (4,716, 37.5%, 45 host listings, 1.79 km from the CBD), W4 (3,249, 2.9%, 2,364 days since review, zero available days) and W5 (1,201, 37.8%, accommodates 10, 17.55 km from the CBD).

## 5. Limitations and Improvement Opportunities

*Preprocessing* — Price imputation assigns the training-set median of \$245.00 to 6,553 missing rows, and 93.35% of those are unavailable listings (`availability_365 = 0`). Separately, `amenity_count` treats every amenity as equally important, so a Wi-Fi router and a pool contribute the same amount. The imputed `days_since_last_review` value of 4,594 becomes an outlier once standardised for k-NN; modelling that group through `has_review` alone, rather than filling a numeric value, was not tried.

*Availability interpretation.* An `availability_365` of 0 is treated as an unavailable listing, but it could also indicate a fully booked calendar or dates the host has blocked rather than a delisted property.

*Correlation* — Observations are not independent. The 21,251 reviewed listings come from 11,474 hosts, and a host's status is the same across all of their listings, so the effective sample is smaller than the row count suggests. At this size almost any coefficient would be significant, so the interpretation rests on effect size rather than on significance.

*MI depends on the number of bins.* For `days_since_last_review` with `availability_365`, MI rises from 0.1695 at five bins to 0.3605 at twenty, an increase of 113%, while NMI moves only from 0.0799 to

0.0896, or 12%. Comparisons between pairs therefore use NMI.

*Supervised Learning* — The classes are imbalanced. Recall on the superhost class is the weak point, at 0.664 for the tree and 0.560 for k-NN.

*Only max\_depth was tuned.* For decision trees, max\_depth is set to 10 and min\_samples\_leaf is left at its default of 1, so a leaf can hold a single listing. Nothing other than depth constrains how finely the tree splits, and jointly tuning min\_samples\_leaf was not attempted.

*The two feature groups are different sizes.* Operational features number eight against three fundamentals, so part of the gap between them may reflect the larger set holding more information rather than that family being more informative. Two further alternatives were not implemented: modelling at the host level, with one row for each of the 14,113 hosts, would remove within-host dependence, and adding the official criteria back in as an upper-bound reference was also left undone.

*Feature Selection* — Continuous features are favoured. A continuous feature offers more possible split points, so it has more chances to reduce impurity. For instance, dist\_cbd ranks 5th by embedded importance at 0.0278 while the filter places it 8th at MI 0.0340.

*Third place is very close.* availability\_365 and amenity\_count differ by only 0.0010 on mutual information (0.0759 versus 0.0749); MI is produced by a nearest-neighbour estimator rather than by folds, so changing the random seed could reverse this order.

*PCA & Clustering* — Much of the variance is unexplained. The first three components carry 69.4%, leaving 30.6% that none of the interpretations above covers, and PC1 alone accounts for 32.1%.

*The result depends on scaling.* Standardising features before PCA is already equivalent to running PCA on the correlation matrix, so that choice does not change the components; the log1p transformation is what matters, because without it long-tailed variables such as host\_listings\_count (maximum 667) and days\_since\_last\_review (maximum 4,593) would dominate the leading components, and the cluster boundaries would move with them.

*k was not selected by a formal test.* k=5 is read from the bend in the elbow curve: SSE falls for every extra cluster, so there is no minimum to look for, and the values for k=4 and k=5 are 72,991.5 and 64,211.4, a difference of about 12%. The silhouette scores for k=4 and k=5 differ by only 0.0005 (0.2227 versus 0.2232), so the advantage of k=5 is small. A gap statistic, or a direct comparison of the k=4 and k=5 cluster profiles, was not carried out.

## 6. Conclusion

The status is partly inferable from the listing record, and the inferable part is operational rather than fundamental. Both feature-selection methods rank days\_since\_last\_review and host\_listings\_count first and second, and the subset experiments settle the comparison: a decision tree on fundamentals alone reaches 0.166 CV F1 against 0.548 for operational features, while on the test set the values are 0.196 and 0.639, with the interval for the difference [0.3813, 0.5046] excluding zero; all 11 features reach 0.5745 CV and 0.6319 test F1 against a zero baseline. PCA and clustering support this conclusion, with PC1 accounting for 32.1% of the variance and K-Means isolating 4,689 low-activity listings at a 6.8% superhost rate. The status can be inferred only in part: test F1 is 0.632 and recall 0.664, finding 1,047 of 1,577 superhost listings while missing 530, and 118 of the 124 superhost listings inactive for over a year remain undetected.

## Generative AI statement

Throughout this project, we utilized Anthropic’s Claude as a generative AI assistance tool. In coding, it was employed to help write and refine analytical scripts (including permutation feature importance, ablation studies, decision tree path analysis, and clustering evaluation metrics), consolidate multiple notebooks into the final submission version, fix a bug in model selection, draft the README, and review code. For the report and presentation, Claude was used solely to review drafts, cross-check numerical consistency across sections, identify errors, and assist in designing the presentation slides, while all report text was authored by team members. All experimental results were executed by the team members locally, every figure was verified against the output files, and the team assumes full responsibility for the entire submission.

## References

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